

MANSOOR AHMED RIND

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PROFESSIONAL SUMMARY

Bachelor's in electrical engineering student at the National University of Sciences and Technology (NUST), Islamabad, Pakistan, with specialized expertise in UAV Electronics and Automation. Highly skilled in robotics systems integration, automation, embedded systems, and full-stack web development. Proven experience in research, development, and deployment of innovative technological solutions, with strong leadership in technical teams and student organizations.

EDUCATION

National University of Sciences and Technology (NUST)

Islamabad, Pakistan

Bachelor of Engineering — Electrical & Electronics Engineering | Sep 2023 – Aug 2027

- Honors: Gold Medalist.

University of California (UCI)

Irvine, California

Non-Credit Specialization — Internet of Things | Coursera

- Specialization: Programming of Internet of things

University of Naples Federico II

Naples, Italy

Non-Credit Specialization — Autonomous Vehicle Engineering

- Specialization: Autonomous Vehicle Engineering

Packt

Non-Credit Specialization — Mastering ROS 2 for Robotics Programming

- Specialization: Mastering ROS 2 for Robotics Programming

TECHNICAL SKILLS

- **Robotics & Navigation:** Autonomous UAV Navigation, Mission Planning, Swarm Robotics, Precision Dynamic/Static Landing, GPS-Denied Navigation, Motion Planning.
- **Software & Frameworks:** ROS, ROS2, MAVROS, PyMAVLink, Gazebo, ArduPilot, PX4, Mission Planner, QGroundControl, MAVProxy.
- **Programming:** Python, C++, OpenCV, YOLO/YOLOv8, HTML, CSS, JavaScript.
- **Embedded & Hardware:** Pixhawk, Arduino, ESP32, Raspberry Pi, UAV Electronics, Flight-Stack Integration.
- **Systems:** Ubuntu/Linux, Kali Linux, Raspbian OS.

RESEARCH & PUBLICATIONS

A Layered ROS 2/Nav2 Architecture for Fail-Safe Indoor UAV Navigation with Deterministic Dynamic-Obstacle Braking

[View publication](#)

FINAL YEAR PROJECT

Autonomous UAV Localization and Mapping in GPS-Denied Environment Using V-SLAM

Sep 2026 – Present

- Advisor: [Dr. Muhammad Moazzam Ali](#)
- Co-Advisor: [Muhammad Saad Zia](#)
- Developing a visual SLAM (V-SLAM) based localization and mapping pipeline for autonomous UAV operation in GPS-denied indoor/outdoor environments.

ENGINEERING & RESEARCH EXPERIENCE

Team AeroMavericks

Islamabad, Pakistan

Founder & Team Captain | Nov 2025 – Present

- Founded and lead a student engineering team developing autonomous UAV systems, overseeing electronics design, embedded programming, system integration and flight-ready prototypes.
- Led an autonomous disaster-relief UAV for National Aerothon '25, integrating electronics, computer vision, GPS navigation and precision-landing algorithms for medical sample collection.

- Achieved 3rd Overall and the “Swift Wing” title after completing autonomous missions simulating hazardous medical environments.

INTELCY IT Solutions

Islamabad, Pakistan

Drone Swarm Engineer | Feb 2026 – Jun 2026

- Worked on drone swarm engineering and multi-UAV autonomy workflows, contributing to distributed coordination and scalable aerial robotics systems.

CSN Lab, SEECS — NUST

Islamabad, Pakistan

Research Student — UAV Autonomy & Swarm Systems | Jun 2025 – Jan 2026

- Engineered precision dynamic and static landing workflows for autonomous UAV missions.
- Implemented decentralized drone-to-drone communication using radio telemetry, enabling swarm coordination independent of a Ground Control Station.
- Developed and tested multi-UAV swarm coordination algorithms in Gazebo and validated autonomy workflows in real-time flight scenarios.
- Integrated onboard computer vision and autonomy pipelines for GPS-denied navigation, distributed control and intelligent aerial systems.

Team Vitesse — Teknofest Turkey

Islamabad, Pakistan

Vice-Captain & Lead Electronics and Automation | Feb 2025 – Sep 2025

- Directed UAV electronics and automation for a dynamic landing system on moving Unmanned Surface Vehicles (USVs).
- Coordinated cross-functional engineering efforts and mission-critical system integration, helping the team become a finalist among 160+ international teams.

Fiverr

Remote

Freelance Robotics & Automation Consultant | Nov 2025 – Present

- Deliver UAV autonomy, embedded systems and intelligent automation solutions for international clients, covering system design, implementation, debugging and technical documentation.

Aerial Robotics Lab, SINES NUST

Islamabad, Pakistan

Robotics Engineer Intern | Jul 2024 – Sep 2024

- Developed and tested UAV autonomy workflows using ROS and MAVROS for GPS-based missions and ArduPilot operations.
- Built an ML-enabled pothole-detection UAV system with onboard processing and server-side data transfer.

Precision Newsletters by One International Services (Pvt.) Ltd

Remote

Web Developer | Sep 2024 – Dec 2024

- Designed and developed a dynamic, user-centric website to enhance brand presence.

KoreaEHT.Co

Remote

Summer Intern | June 2024 – August 2024

- Conducted detailed research on Electric Heating Systems and EHT cables.

National School & College System

Islamabad

Robotics Workshop Instructor

- Delivered engaging sessions on the fundamentals of robotics and future applications.

SELECTED TECHNICAL PROJECTS

- **Custom Build Ground Control Station for UAV Swarms:** Designed a centralized, MAVLink-integrated ground control interface for real-time telemetry monitoring, simultaneous mission planning, and decentralized command of multi-drone formations.
- **Autonomous Kamikaze UAVs for Warfare:** Engineered a vision-guided loitering munition prototype utilizing onboard object detection, locking and terminal guidance algorithms for dynamic target interception in defense scenarios.
- **Autonomous Interceptor UAV:** Developed an autonomous UAV platform for dynamic tracking and response applications.
- **UAV Orthomosaic Mapping:** Developed a ROS-based orthomosaic mapping workflow for aerial mapping applications.
- **GPS-Denied Navigation:** Developed vision-based velocity-control workflows for autonomous UAV operation without GNSS.
- **Multi-UAV Swarm Formation:** Developed decentralized inter-UAV coordination and swarm formation workflows.
- **YOLOv8 Road Crack Detection:** Developed a computer-vision pipeline for intelligent road-conditioning inspection.
- **VTOL UAV Prototype:** Worked on autonomous VTOL system development and integration.
- **ROS/Gazebo Autonomous Flight:** Simulated ArduCopter autonomous missions using ROS and Gazebo.

LEADERSHIP & LARGE-SCALE OPERATIONS

NUST Placement Office — President NUST Young Student Exchange Program / Office Representative NUST Internship Program for International Students '26

Jun 2026 – Aug 2026

- Supported NUST international internship operations for 51 international interns from 15+ countries; led registration, onboarding and operational workflows.

NUST Career Connect 2026 — Coordinator

Dec 2025 – Feb 2026

- Coordinated security, finances, registration and logistics for a career fair involving 150+ companies; managed registration workflows for 130+ companies.

NUST Digital Club — General Secretary / VP Operations

Sep 2025 – Jun 2026

- Led Administration, Events, Sponsorship, Marketing and Security/Protocols teams, coordinating cross-functional operations.

NUST Orientation — Director, Liaison & Registration

Jul 2025 – Sep 2025

- Directed registration and onboarding workflows for 3,000+ incoming students, coordinating participants, faculty and organizing committees.

CONNEX'24 National HR Conference by PSTD — Student Volunteer Lead

CAREC by Asian Development Bank — Student Volunteer

Conference of Parties (COP 24) by UNICEF x Youth4Pak — Student Volunteer

Gender Equity and Climate Change Conference by German Cooperation x NUST — Student Volunteer

COMPETITIONS & RECOGNITION

- 3rd Overall — National Aerothon '25 | Swift Wing title | Autonomous Disaster-Relief UAV
- Finalist — Teknofest Turkey 2025 | Dynamic UAV Landing on Moving USV | 160+ international teams
- Participant — IMechE UAS Challenge 2025 | Disaster Relief UAV
- Ranked 112/200 — International Design, Build & Fly Competition 2025
- Finalist — National Design, Build & Fly Competition 2024
- Participant — National Engineering Robotics Competition 2024 & 2025
- Finalist — Teknofest Turkey 2024 | Anti Drone UAV | 160+ international teams

CERTIFICATIONS & LANGUAGES

Certifications: UAS Remote Pilot Open Category — A1+A3 | Introduction to Nephio LFS179

Languages: English (Professional Working) | Urdu (Native/Bilingual) | Sindhi (Native/Bilingual)